INCH-POUND

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31 January 2005
SUPERSEDING
MIL-STD-801B
12 August 2002

DEPARTMENT OF DEFENSE
TEST METHOD STANDARD

INSPECTION AND ACCEPTANCE STANDARDS FOR FUEL CELLS AND FITTINGS

AMSC N/A

FSC 1560

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This test method standard establishes:

a. A classification of commonly occurring defects in fuel cells and fuel cell fittings.

b. Standards for normal finish operations.

c. Standards for acceptance limits of rework on new fuel cells in the manufacturer's plant.

d. Limits of acceptable conditions requiring no rework.

e. Inspection criteria for acceptance determination of fuel cells that have been subjected to quality control stand test or dissection tests in accordance with MIL-T-5578, MIL-T-6396, and MIL-DTL-27422.

3. Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or via email to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

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1. SCOPE

1.1 Scope. This standard establishes a test method standard for fuel cells and fuel cell fittings. This standard applies to all fuel cells manufactured in accordance with MIL-T-5578, MIL-T-6396 (except type I), and MIL-DTL-27422.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-5578	-	Tank, Fuel, Aircraft, Self-Sealing
MIL-T-6396	-	Tanks, Aircraft Propulsion Fluid System, Internal, Removable, Non-self-sealing (Inactive for New Design)
MIL-DTL-27422	-	Tank, Fuel, Crash-Resistant, Ballistic-Tolerant, Aircraft

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ-Z1.4	- Procedures, Sampling and Tables for Inspection by Attributes (DoD adopted)

(Copies are available from www.asq.org or the American Society for Quality (ASQ), P.O. Box 3005, Milwaukee, WI 53201-3005.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D751 - Fabrics, Coated (DoD adopted)

(Copies are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Approved rework procedures. Approved rework procedures are those rework procedures that have been approved by the fuel cell design authority.

3.2 Back rind. Back rind is a surface blemish defect caused by mold flash material becoming folded inside a fitting cavity.

3.3 Baffle shoes. Baffle shoes are fabric straps usually having holes protected by grommets. These straps are attached to the liner of the fuel cell for the purpose of securing the internal baffles.

3.4 Bleeder cords. Bleeder cords are cords which are built between the plies for the purpose of evacuating solvents and trapped air from between plies of a fuel cell.

3.5 Bleeder patch. A bleeder patch is a patch on the outside of a fuel cell, which covers the cut ends of bleeder cords.

3.6 Blister. A blister is an area of no adhesion between layers of the fuel cell wall.

3.7 Crash resistant fuel cells. The column heading “crash resistant fuel cells”, which appears in tables I, II, and III applies to both self-sealing and non-self-sealing crash resistant fuel cells.

3.8 Delamination. A delamination is an area of no adhesion between plies of the fuel cell wall. A delamination is not an area in which adhesion between plies is intermittent, such as where the fabric inner liner adheres primarily to the high points produced by the coarse weave of a reinforcing layer of fabric and does not adhere to the low points (valleys). These areas are essentially unpressurized and do not prevent the cell from satisfactorily holding fuel.

3.9 Design failure. Failures caused by a physical breakdown of the fuel cell or attach points as a result of imposed loads, which the cell was designed to withstand.

3.10 Deterioration of construction or construction failure. A failure caused by action of the test fluid on any ply or coat of cement or barrier in the fuel cell.

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3.11 Dissection test. A destructive test in which the fuel cell is sectioned to permit critical examination of the interstitial areas of the fuel cell (see MIL-T-5578, MIL-T-6396, and MIL-DTL-27422).

3.12 Fitting leakage. Leakage from a fuel cell fitting.

3.13 Fried, scarred, or blown condition. An area in the liner material, which has become sponge like, where solvent laden adhesive has "blown" during the vulcanization procedure.

3.14 Hanger straps. Straps applied to the outside surface of fuel cells for the purpose of handling cells or securing cells in the cavities.

3.15 Integral baffle. An integral baffle is a rubber coated fabric component, which is vulcanized into the tank wall construction as a part of the building process.

3.16 Looseness. Area of non-adhesion.

3.17 Lumps and craters. Depressions or craters caused by scuffing of uncured gum stocks or inclusion of cement lumps or foreign material.

3.18 Manufacturing damage. Damage incurred while the fuel cell is in the process of manufacture.

3.19 Manufacturing defect. A defect caused by the fuel cell not being fabricated in accordance with applicable drawings and specifications.

3.20 Multiple construction fuel cell. A fuel cell which consists of more than one basic construction.

3.21 Rework damage. Damage incurred during rework or finishing operations.

3.22 Source inspector. The quality control inspector or engineer who is directly responsible for the procuring activity quality control function.

3.23 Stand test. A stand test is a static test in which the test cell is filled with fuel or test fluid for a given period of time and then examined for evidence of leakage or material deterioration (see MIL-T-5578, MIL-T-6396, and MIL-DTL-27422).

3.24 Step off. The edge of a lap of one or more plies of materials. The edge of discontinued plies in a multiple construction fuel cell.

3.25 Total effective bond. Amount of bond area in fitting flange or lap splice whether or not bond is continuous.

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4. GENERAL REQUIREMENTS

4.1 Inspection requirements. The inspection requirements described herein are not intended to supersede or delete any existing quality control standards. It is expected that the manufacturer or procuring activity shall conduct further tests and have other detailed requirements in excess of those specified herein.

4.2 Results. Report of test results and inspections specified in this standard are in accordance with the applicable contract requirements.

5. DETAILED REQUIREMENTS

5.1 Classification of defects. Defects are considered with regard to character and extent. Defects are classified and limited to fall within one or more of the following classes as specified in ANSI/ASQ-Z1.4.

a. Critical. A critical defect is one that judgment and experience indicate could result in hazardous or unsafe conditions for individuals using or maintaining the product; or for major end-item units of which the product is a component, such as ships, aircraft, or tanks; a defect that could prevent performance of their tactical function.

b. Major. A major defect is a defect, other than critical, that could result in failure or materially reduce the usability of the unit or product for its intended purpose.

c. Minor. A minor defect is one that does not materially reduce the usability of the unit or product for its intended purpose or is a departure from established standards having no significant bearing on the effective use or operation of the unit.

5.1.1 Unlisted defects. Unlisted defects, when deemed by the Inspector to be such as to adversely affect the serviceability or strength of the fuel cell or fitting, are classified in accordance with the above criteria.

5.2 Use of tables. (See table V for summary)

5.2.1 Table I. Corrections of discrepancies listed in table I are considered to be normal finishing operations and are not counted as defects when properly reworked prior to the time the fuel cell (or fitting) is submitted for acceptance or if the rework is accomplished prior to initiation of stand tests. An "X" in one or more of the seven columns below “Types of fuel cells” indicates the type of cell(s) or fitting(s), or the portion of the cell (interior, exterior) to which a particular inspection is considered applicable.

5.2.2 Table II. This table presents acceptance standards for various defects that may occur on fuel cells that are submitted by the manufacturer for acceptance by the procuring activity. This

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table is also used for inspection of stand or dissection tested fuel cells (see 5.2.3). The seven columns below “Type of fuel cells” indicate the applicability of each defect with regard to type of cell, fitting, or location of defect (interior, exterior).

5.2.3 Table III. This table lists inspections and defects that are unique to stand tested fuel cells. For stand tested cells, the inspections listed in table III are conducted in addition to the inspections in table II. Stand tests are normally conducted in fixtures lined with indicator paper. The "stains" referred to in table III are the stains showing on the indicator paper when cells are defective. If an alternate method of leak detection is used for stand tests, equivalent criteria are used for defect classification. Leaks that cannot be attributed to physical damage or shown to be a unique case are classified as a critical defect.

5.2.4 Table IV. Table IV lists additional defects that are applicable to dissection tested fuel cells. These inspections are conducted on dissection tested cells in addition to the inspections in table II. Item 6 in table IV is applicable to replacement fuel cell fittings that are subjected to dissection tests on a sampling basis as well as to dissection tested cells.

5.2.5 Table V. Table V summarizes the contents of 5.2 and 5.3.

5.3 Action to be taken on identified defects. (See table V for a summary of these actions.)

5.3.1 Minor defects. Discrepancies classified as minor defects are considered acceptable without rework provided they do not exceed the following limits:

a. Fuel cell interior - One defect per 10 square feet of total cell area. Where an accumulation of minor defects is not greater than the limits specified in the table, they will be considered one defect. For example, an accumulation of five 1/4-inch blisters in any one 5-foot length of splice (table II, defect 1a) is counted as one minor defect.

b. Fuel cell exterior - Not greater than limits specified in table.

c. Installed fuel cell fittings - Minor defects in installed fittings are counted and included when determining acceptability of fuel cells without rework in accordance with 5.3.1, a and b.

d. Replacement fuel cell fittings - Replacement fittings submitted for acceptance are considered as acceptable without rework if minor defects are not greater than the following:

(1) Fittings with bolt circle of 6 inches or less - two minor defects permitted.

(2) Fittings with bolt circles larger than 6 inches - three minor defects permitted.

For non-circular fittings, consider bolt circle to be the largest dimension of the fitting. Minor defects greater than the criteria listed above are considered a major defect and reworked prior to acceptance.

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5.3.2 Major defects. Major defects in fuel cells or replacement fuel cell fittings submitted for acceptance under a production contract are corrected using an approved rework procedure prior to acceptance. Major defects detected in cells during stand or dissection tests or in separate fittings subjected to destructive sampling tests are analyzed to determine cause. If the defect can be shown to be a unique case, the lot represented by the defective item may be accepted. If not unique, all like defects shall be reworked using the approved rework procedure prior to acceptance of the lot and appropriate alterations must be made in the manufacturing process to prevent repetition of the defect.

5.3.3 Critical defects. Critical defects in fuel cells or replacement fuel cell fittings are cause for rejection of the cell. The cell may be deliverable if the manufacturer can devise a special rework technique along with a test procedure, both of which are acceptable to the procuring activity. When a critical defect is identified in cells subjected to stand or dissection tests, the procuring activity is notified immediately. If a critical defect is identified during the production sampling's test, production is stopped until the problem is resolved and a course of action established that is mutually satisfactory to the cell manufacturer and the procuring activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard classifies common defects in fuel cells and fuel cell fittings. This standard establishes the requirements for normal finish operations and sets acceptance limits or rework on new cells. These finish operations and acceptance limits are intended for military aircraft fuel tanks that must pass ballistic tests.

6.2 Acquisition requirements. Acquisition documents should specify the title, number, and date of the specifications cited.

6.3 Supersession note. When referenced in a specification or government acquisition document, MIL-STD-801B governed the inspection of production fuel cells in the manufacturer's and prime contractor's plants or at government facilities to the extent specified in the contract under which the cells were acquired. The following documents were superseded by MIL-STD-801: ANA Bulletin 107, "Inspection Standards for Stand and Dissection Tested Self-Sealing Fuel and Oil Cells;" ANA Bulletin 112, "Acceptance Standards for Self-Sealing Fuel and Oil Cells; and ANA Bulletin 435, "Inspection Standards for Stand and Dissection Tested Non-Self-Sealing Type Cells."

6.4 Subject term (key word) listing.

Aircraft
Bladder
Classification
Crash resistant
Critical
Defects
Finish operations
Fuel

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Leakage
Rework
Self-sealing
Stand test

6.5 Changes from previous issue. The margins of this standard's tables are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE I. Fuel cell finishing operations. 1/

No.	Defects	Types of Fuel Cells						
		Bladder Cells	Self Sealing Cells	Cell Exterior	Cell Interior	Replacement Fittings	Installed Fittings	Crash Resistant Cells
I	LOOSE CORNER PATCH	X	X	X	X	X	-	X
2	BLISTERED CHAFING STRIPS	X	X	X	X	-	-	X
3	LOOSE BAFFLE SHOES/INTEGRAL BAFFLE	X	X	-	X	-	-	X
4	LOOSE LINER AT THROAT OF FITTING THAT CAN BE REWORKED BY TRIMMING, RETAINING MINIMUM BOND PERMITTED BY APPLICABLE SPECIFICATION	X	X	-	X	-	X	X
5	LOOSE LINER OR OUTER PLY LAP THAT CAN BE REWORKED BY TRIMMING OR USE OF ADHESIVES MAINTAINING MINIMUM BOND PERMITTED BY APPLICABLE SPECIFICATION	X	X	X	X	-	-	X
6	EDGE LOOSENESS ON REINFORCEMENT, ATTACHING STRAPS, CHAFING STRIPS, TABS, ETC.	X	X	X	X	-	-	X
7	EXPOSED FABRIC EDGES AFTER CURE, PROVIDED FABRIC IS NOT DAMAGED.	X	X	X	X	-	-	X
8	BLISTERS BETWEEN LINER OR OUTER PLY AND FITTING FLANGES	X	X	X	X	-	X	X
9	DAMAGED GROMMETS IN ACCESSORIES	X	X	X	X	-	-	X
10	DAMAGED COATING ON METAL, RUBBER, OR COMPOSITE ACCESSORIES	X	X	X	X	-	-	X
11	IMPROPER OR LACK OF IDENTIFICATION MARKING, EXCEPT REPLACEMENT OF CEMENTED LABELS	X	X	X	X	-	-	X
12	SKIM COAT BLISTERS	X	X	X	-	-	-	X

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TABLE I. Fuel cell finishing operations – Continued. 1/

No	Defects	Types of Fuel Cells						
		Bladder Cells	Self Sealing Cells	Cell Exterior	Cell Interior	Replacement Fittings	Installed Fittings	Crash Resistant Cells
13	SKIM COAT OFF OUTER PLY, PROVIDED CORDS OR FABRIC NOT DAMAGED	X	X	X	-	-	-	X
14	MISSING BLEEDER PATCH	X	X	X	-	-	-	X
15	MISSING OR MISLOCATED TRIM TAPE	X	X	X	-	-	-	X
16	RUST OR DIRT IN THREADS OR DEFECTIVE THREADED INSERTS	-	-	-	-	X	X	-
17	METAL FINISH DAMAGE	-	-	-	-	X	X	-
18	BENT FITTINGS THAT CAN BE STRAIGHTENED WITHIN STRESS LIMITATIONS	-	-	-	-	X	X	-
19	FLASH ON FITTING FLANGE TIPS, INSIDE DIAMETERS, THROUGH HOLES, ETC.	-	-	-	-	X	X	-
20	TEARS IN FITTING FLANGES THAT CAN BE TRIMMED WITHOUT VIOLATING APPLICABLE SPECIFICATION MINIMUM FLANGE LENGTH	X	X	X	X	X	X	X
21	DEFECTS SUCH AS MOLDING MARKS, CUTS, OR DEPRESSIONS THAT CAN BE CORRECTED BY BUFFING AND STILL MAINTAIN SPECIFICATION TOLERANCES	X	X	X	X	X	X	X
22	EXPOSED FABRIC CORDS THAT CAN BE CORRECTED BY SEAL COATING PROVIDED FABRIC IS NOT DAMAGED	X	X	X	X	X	X	X

1/ Corrections of the items listed in this table are normal finishing operations and are not counted as a defect or rework when completed. See table II for defects that do not require rework.

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TABLE II. Acceptance standards. 1/

Types of Fuel Cells							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	X	X	X	-	-	-	1. AREAS OF NONADHESION OR BLISTERS BETWEEN LINER LAPS			
							a. 1/4 inch maximum dimension (half inch for crash resistant cells) - Average 1 per 5 linear ft. of splice with maximum of 5 in any one 5 ft. length of splice	X	-	-
							b. Areas in excess of 1.a.	-	X	-
X	X	X	X	-	-	-	2. BLISTERS BETWEEN PLYS OR BETWEEN INNER LINER AND SEALANT			
							a. 1 inch maximum dimension	X	-	-
							b. Blisters between ply in excess of 2.a.	-	X	-
X	X	X	X	X	X	-	3. .BLISTERS BETWEEN FITTING FLANGE AND ADJACENT PLY			
							a. 1/4 inch maximum dimension – Maximum of 1 in any 1 linear ft. of flange, or maximum of 1 per fitting when flange is less than 1 linear ft.	X	-	-
							b. In excess of 3.a.	-	X	-
X	X	X	X	X	-	-	4. BLISTERS BETWEEN ATTACHMENTS AND OUTER OR INNER PLY			
							a. Not exceeding 15 percent looseness provided a continuous bond at least 1/4 inch wide is maintained around edge except at step-off where item 8 applies.	X	-	-
							b. In excess of 4.a.	-	X	-

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TABLE II. Acceptance standards - Continued. 1/

TYPES OF FUEL CELLS							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	X	X	X	X	-	-	5. CHANNELS BETWEEN PLIES AT BURIED EDGE OF LAP SPLICES AND IN TRANSITION AREA, ENTIRE LENGTH OF SPLICE			
							a. In excess of 1/4 inch in width	-	X	-
							b. In excess of 3/8 inch in width for crash resistant cells	-	X	-
X	X	X	X	X	X	-	6. CHANNEL AROUND ENTIRE OUTER EDGE OF FITTING FLANGE			
							a. In excess of 1/4 inch in width	-	X	-
							b. In excess of 1/2 inch in width for crash resistant cells	X	-	-
X	X	X	X	X	X	X	7. CHANNEL AROUND ENTIRE THROAT OF FITTING FLANGE			
							a. Up to 1/4 inch in width	X	-	-
							b. In excess of 7.a.	-	X	-
							c. In excess of 1/2 inch in width for crash resistant cells	-	X	-
X	X	X	X	X	-	-	8. CHANNELS AT FABRIC COMPONENTS			
							a. 1/4 inch wide maximum dimension - Maximum of 1 in any 1 linear ft	X	-	-
							b. In excess of 8.a.	-	X	-

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TABLE II. Acceptance standards - Continued. 1/

Types of Fuel Cells							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	X	X	X	-	-	-	9. OPEN END CHANNELS IN 3 PLY-LINER OVERLAPS OR TAILORED CORNERS			
							a. Less than 1/4 inch wide by 3 inches long and less than 1 inch bond maintained between end of channel and barrier (N/A if 1 inch bond is maintained)	X	-	-
							b. In excess of 1/4 inch wide by 3 inches long and less than 1 inch bond maintained between end of channel and barrier	-	X	-
X	-	X	X	-	-	-	10. FRIED OR SCARRED CONDITION OR THINNING OUT OF GUM LINERS IN BLADDER CELLS			
							a. Less than 1/2 of specified thickness	X	-	-
							b. In excess of 1/2 of specified thickness	-	X	-
X	X	-	X	-	-	-	11. FRIED OR SCARRED CONDITION OR THINNING OUT OF GUM INNER LINER IN SELF-SEALING CELLS - In excess of 1/2 of specified liner thickness (less than 1/3 – no defect)	-	X	-
X	X	X	X	-	-	-	12. CUTS OR HOLES IN INNER LINER (gum or fabric inner liner) 2/	-	-	X
X	X	X	X	-	-	-	13. LUMPS, CRATERS OR FOREIGN MATERIALS IN FUEL CELL GUM INNER LINERS			
							a. Up to 1/2 of specified liner thickness	X	-	-
							b. In excess of 1/2 of specified liner thickness	-	X	-

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TABLE II. Acceptance standards - Continued. 1/

Types of Fuel Cells							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	X	X	X	X	-	-	14. SEPARATION OF CEMENTED COMPONENTS (CHAFING, STRIPS, TABS, ETC.)			
							a. Not exceeding 15 percent looseness provided a continuous bond at least 1/4 inch wide is maintained around edge except at step-off where item 8 applies.	X	-	-
							b. In excess of 14.a.	-	X	-
X	X	X	X	X	-	-	15. DELAMINATION OR SEPARATION BETWEEN ANY TWO PLIES OR BETWEEN LINER AND ADJACENT PLY			
							a. 1 inch maximum dimension; not to exceed average of 1 per 5 sq. ft area; maximum of 5 inches any one 5 sq. ft area; minimum of 6 inches solid bond between delaminations	X	-	-
							b. In excess of 15.a.	-	X	-
X	X	X	-	X	-	-	16. EXTERNAL DIMENSIONS OF CELL OUT OF TOLERANCE	-	-	X
X	X	X	X	X	X	-	17. FITTING MISLOCATED OR FITTING ROTATED BEYOND SPECIFIED TOLERANCE	-	-	X
X	X	X	X	X	-	-	18. LAP SPLICE EDGE LOOSENESS			
							a. Up to 1/8 inch in width and 3 inches long; no more than 1 per 5 linear ft, and rework can be made by trimming	X	-	-
							b. In excess of 18.a. or if rework cannot be made by trimming	-	X	-
X	X	X	-	X	-	-	19. OUTER PLY CUTS OR SPLITS PARALLEL TO CORDS IF CORDS ARE NOT DAMAGED	X	-	-
X	X	X	X	X	X	-	20. LOOSENESS UNDER HANGER FITTINGS INCORPORATING METAL OR FIBER PLATES OR METAL RINGS, IN EXCESS OF THE CONTACT SURFACE AREA OF THE PLATES AND RINGS AND UP TO 1/4 INCH BEYOND OUTER EDGE OF PLATE OR RING		X	-

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TABLE II. Acceptance standards - Continued. 1/

Types of Fuel Cells							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	-	-	-	-	-	X	21. BLISTERS IN FLANGE AREA OF REPLACEMENT FITTINGS	-	X	-
X	X	X	X	X	X	X	22. MOLD MARKS IN FLANGE: AREA OF FITTING			
X	X	X	X	X	X	X	a. No interference with design characteristics of fitting	X	-	-
							b. Sufficient to cause interference	-	X	-
X	X	X	X	X	X	X	23. FLOW CRACKS, LAMINATIONS. POROUS AREAS OR BACK RINDS IN EXCESS OF 1/2 BODY THICKNESS OF FITTING FLANGE AT LOCATION OF DEFECT	-	X	-
X	X	X	X	X	X	X	24. FOREIGN MATERIAL IN FITTING FLANGE			
							a. 1/32 inch or less	X	-	-
							b. Over 1/32 inch but less than 1/2 body thickness at defect	-	X	-
X	X	X	X	X	X	X	25. TEARS IN FITTING FLANGE			
							a. If fairing can be accomplished while still maintaining specified minimum flange width	X	-	-
							b. In excess of 25.a.	-	X	-
X	X	X	X	X	X	X	26. CUTS OR NICKS IN FITTING FLANGE			
							a. 1/32 inch or less	X	-	-
							b. Over 1/32 inch but less than 1/2 body thickness at defect	-	X	-
							c. Greater than 1/2 body thickness	-	-	X

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TABLE II. Acceptance standards - Continued. 1/

Types of Fuel Cells							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	X	X	X	X	X	X	27. EXPOSED FABRIC IN FITTING FLANGE			
							a. With no damaged cords. Cords exposed at edge of fitting flange are permitted	X	-	-
							b. With damaged cords	-	X	-
X	X	X	X	X	X	X	28. BLISTERS IN FITTING FILET			
							a. Up to 1/8 inch	X	-	-
							b. In excess of 1/8 inch	-	X	-
X	X	X	X	X	X	X	29. CUTS OR CRACKS IN FITTING FILLETS	X	-	-
X	X	X	X	X	X	X	30. EDGE LOOSENESS BETWEEN FITTING FILLET AND FABRIC	X	-	-
X	X	X	X	X	X	X	31. EDGE LOOSENESS BETWEEN FITTING FILLET AND METAL	X	-	-
X	X	X	X	X	X	X	32. SEALING SURFACE OF FITTING ROUGHER THAN SPECIFICATION LIMITS	-	X	-
X	X	X	X	X	X	X	33. FITTING CRITICAL FLAT AREA FINISH, OTHER THAN SEALING SURFACE, ROUGHER THAN SPECIFICATION LIMITS	-	X	-
X	X	X	X	X	X	X	34. SCRATCHES, PITS OR MARS IN SEALING SURFACE BEYOND LIMITS OF SPECIFICATION	-	-	X
X	X	X	X	X	X	X	35. MOLDING FLASH ON SEALING SURFACE OF FITTING	-	-	X
X	X	X	X	X	X	X	36. FAULTY DOME NUT	-	-	X

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TABLE II. Acceptance standards - Continued. 1/

Types of Fuel Cells							Defects	Defect classes		
Crash Resistant Cells	Self Sealing Cells	Bladder Cells	Cell Interior	Cell Exterior	Installed Fittings	Replacement Fittings		Minor	Major	Critical
X	X	X	X	X	X	X	37. THREAD DIMENSIONS OUT OF TOLERANCE	-	X	-
X	X	X	X	X	X	X	38. CORROSION, DIRT, METAL FINISH DAMAGE OR LACK OF SPECIFIED PROTECTIVE COATING	-	X	-
X	X	X	X	X	X	X	39. O-RING GROOVE WIDTH OR DEPTH, FITTING FLATNESS, THICKNESS I.D. OR BOLT HOLE DIMENSIONS OUT OF TOLERANCE	-	X	-
X	X	X	X	X	X	X	40. BENT FITTING			
							a. Can be straightened within stress limits of metal	-	X	-
							b. Cannot be straightened within stress limits of metal	-	-	X
X	X	X	-	-	-	X	41. BUFFING THROUGH FABRIC INNER LINER	-	-	X
X	X	X	X	-	-	X	42. ANY CONDITION CAUSING EXPOSED FABRIC; NO DAMAGE OF FABRIC INNER LINER	X	-	-
X	X	X	-	-	-	X	43. FABRIC COATING BLISTERS OF FABRIC INNER LINERS	X	-	-

1/ Unless otherwise specified in the applicable detail specification, a one-inch minimum effective bond shall be maintained. A defect violating the applicable minimum bond requirement shall be classified as a major defect.

2/ This item is not applicable to stand test or dissection test. See tables III & IV.

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TABLE III. Additional criteria for stand tested fuel cells. 1/

Fuel Cells			DEFECTS	DEFECT CLASSES		
CRASH RESISTANT CELLS	SELF SEALING CELLS	BLADDER CELLS		MINOR	MAJOR	CRITICAL
X	X	X	1. Stain or activation due to deterioration of cell 2/	-	-	X
X	X	X	2. Stain or activation due to design failure 2/	-	-	X
X	X	X	3. Stain or activation due to manufacturing defect 2/	-	-	X
X	X	X	4. Stain or activation due to manufacturing damage or rework damage 2/	-	-	X
X	X	X	5. Stain or activation due to mishandling or improper cell 2/	-	-	-
			a. Condition of cell does not preclude complete evaluation of cell	no defect	-	-
			b. Condition precludes complete evaluation	retest	-	-
-	X	-	6. Activation anywhere in the cells	-	X	-

1/ Following stand test, the above inspection standards shall apply in addition to the acceptance standards of table II.

2/ If it is impossible to determine source of stain by thorough engineering and laboratory tests, excluding dissection, a stain of 1 inch maximum diameter or equivalent area is allowable, except self sealing fuel cells, provided there is not greater than one such stain per 150 sq. ft. of cell area.

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TABLE IV. Additional criteria for dissection tested fuel cells. 1/

CRASH RESISTANT CELLS	SELF SEALING CELLS	BLADDER CELLSS	DEFECTS	MINOR	MAJOR	CRITICAL
X	X	X	1. Adhesion of inner liner to sealant or adjacent ply in flat panel areas is less than 6 lb. per inch of width 2/	-	X	-
X	X	X	2. Adhesion of inner liner to chafing strips less than 6 lb. per inch of width 2/	-	X	-
X	X	X	3. Adhesion of fitting flange to inner ply less than 6 lb. per inch of width 2/	-	X	-
X	X	X	4. Adhesion of inside and outside accessories less than design values 2/	-	X	-
X	X	X	5. Dissection of fitting flange indicates construction is not in accordance with approved manufacturing specifications	-	X	-
X	X	X	6. Adhesion of fabric fitting flanges to metal ring less than 150 lb. per inch of width. 2/ 3/	-	X	-

1/ Following dissection test, the above inspection standards shall apply in addition to the acceptance standards of table II.

2/ Test Method: ASTM-D751, Dry Pull at 2 inches per min. This factor also used for sampling tests of replacement fittings.

3/ Failure of fabric flange prior to adhesion failure does not constitute a defect.

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TABLE V. Summary.

PURPOSE OF INSPECTION	APPLICABLE TABLE	ACTION TO BE TAKEN WITH DEFECTIVE ITEMS		
		MINOR DEFECT	MAJOR DEFECT	CRITICAL DEFECT
Production cells submitted for acceptance.	Table II	Cells with minor defects are acceptable without rework provided the defects do not exceed the limits noted in 5.3.1a, b, & c. If these limits are exceeded, they shall be treated as major defects.	All major defects shall be corrected using an approved rework or repair procedure prior to acceptance.	Cells with critical defects shall be rejected. The cell shall be scrapped unless the manufacturer proposes a rework or repair procedure acceptable to the procuring activity.
Replacement production fittings submitted for acceptance	Table II	Fittings with minor defects are acceptable without rework provided the defects do not exceed the limits noted in 5.3.1d. If these limits are exceeded; they shall be treated as major defects.		

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TABLE V. Summary - Continued.

PURPOSE OF INSPECTION	APPLICABLE TABLES	ACTION TO BE TAKEN WITH DEFECTIVE ITEMS		
		MINOR DEFECT	MAJOR DEFECT	CRITICAL DEFECT
Inspection of cells subjected to stand test or dissection test.	Table II & either Table III or Table IV as applicable.	Test results shall be considered satisfactory if minor defects do not exceed limits listed in applicable portion of 5.3.1. If limits are exceeded, they shall be treated as major defects.	When major defects are identified, production items shall be given approval provided it can be shown that this is a unique case. If it is not a unique case, all like defects in production cells shall be reworked or repaired. Lot represented by test is then acceptable. Take appropriate action to prevent repetition of defect.	When a critical defect is identified, test is considered unsatisfactory. Notify prime contractor and procuring activity. All production represented by test shall be inspected to determine if this is a unique case and shall be found free of this defect prior to acceptance. If this is a unique case, test cell shall be reworked or repaired, if possible, utilizing approved procedures. If not a unique case, action shall be taken to determine if defective cells can be reworked or repaired and the defect shall be corrected in future production.
Replacement fittings subjected to dissection test.	Table II & Item 6 in Table IV			

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Review activities:
Army - AT, MI
Navy - SA

Preparing activity:
Navy - AS

(Project 1560-0035)

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